Intoduction

Statistics is concerned with making inferences about populations and population characteristics. Experiments are conducted with results that are subject to chance. The testing of a number of electronic components is an example of a statistical experiment, a term that is used to describe any process by which several chance observations are generated. It is often important to allocate a numerical description to the outcome.

For example, the sample space giving a detailed description of each possible outcome when three electronic components are tested may be written $S = \{NNN, NND, NDN, DNN, NDD, DND, DDN, DDD\}$, where N denotes nondefective and D denotes defective.

One is naturally concerned with the number of defectives that occur. Thus, each point in the sample space will be assigned a numerical value of $0, 1, 2$, or 3. These values are, of course, random quantities determined by the outcome of the experiment. They may be viewed as values assumed by the random variable X, the number of defective items when three electronic components are tested.

A random variable is a variable that associates a real number with each element in the sample space.

Random Variables

Intoduction

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X : (Sample space) $\longrightarrow \mathbb{R}$.

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A random variable is a variable that associates a real number with each element in the sample space.

Example

Two balls are drawn in succession without replacement from an urn containing 4 red balls(R) and 3 black balls(B). The possible outcomes $S = \{RR, RB, BR, BB\}$. Then we define a function X from S to real number $\mathbb{R}$, where X represent the number of red balls for every sample of S

The variable X, representing the number of red balls in a sample of S.
Define $X : S \longrightarrow \mathbb{R}$ by

$$X(s_1) = 2 \qquad X(RR) = 2$$
$$X(s_2) = 1 \qquad X(RB) = 1$$
$$X(s_3) = 1 \qquad X(BR) = 1$$
$$X(s_4) = 0 \qquad X(BB) = 0$$

where $s_i \in S, i = 1, 2, 3, 4$.

Thus the variable X is called a random variable with the values x is $0, 1, 2$.

If a sample space contains a finite number of possibilities or an unending sequence with as many elements as there are whole numbers, it is called a discrete sample space.

If a sample space contains an infinite number of possibilities equal to the number of points on a line segment, it is called a continuous sample space.

A random variable is called a discrete random variable if its set of possible outcomes is countable.

When a random variable can take on values on a continuous scale, it is called a continuous random variable.

Discrete Probability Distributions

A discrete random variable assumes each of its values with a certain probability.

In case of tossing a coin two times, the possible outcomes are $S = \{HH, HT, TH, TT\}$. The variable X, representing the number of heads in $s \in S$. Now, we define by $X : S \longrightarrow \mathbb{R}$ as

$$X(s_1) = 2$$

$$X(s_2) = 1$$

$$X(s_3) = 1$$

$$X(s_4) = 0.$$

Thus the random variable X and the values of x with the probability values of x is given below table.

X	0	1	2
P(x)=P(X=x)	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

The set of ordered pairs $(x, P(x))$ is called the probability mass function, probability function, or probability distribution of the discrete random variable X.

Probability mass function

The set of ordered pairs $(x, P(x))$ is a probability mass function, probability function, or probability distribution of the discrete random variable X if, for each possible outcome x,

- $P(x) \geq 0$
- $\sum_{x} P(x) = 1$.

Example

A shipment of 20 similar laptop computers to a retail outlet contains 3 that are defective. If a school makes a random purchase of 2 of these computers, find the probability distribution for the number of defectives.

Solution

Let X be a random variable whose values x are the possible numbers of defective computers purchased by the school. Now,

- $P(X=0) = \frac{\binom{3}{0}\binom{17}{2}}{\binom{20}{2}} = \frac{68}{95}$
- $P(X=1) = \frac{\binom{3}{1}\binom{17}{1}}{\binom{20}{2}} = \frac{51}{190}$
- $P(X=2) = \frac{\binom{3}{2}\binom{17}{0}}{\binom{20}{2}} = \frac{3}{190}$

Thus, the probability distribution of X is given below table.

X	0	1	2
P(x)=P(X=x)	$\frac{68}{95}$	$\frac{51}{190}$	$\frac{3}{190}$

Continuous Probability Distributions

A continuous random variable has a probability of 0 of assuming exactly any of its values. Consequently, its probability distribution cannot be given in tabular form.

Let us discuss a random variable whose values are the heights of all people over 21 years of age.
Between any two values, say 163.5 and 164.5 centimeters, or even 163.99 and 164.01 centimeters, there are an infinite number of heights, one of which is 164 centimeters.

The probability of selecting a person at random who is exactly 164 centimeters tall and not one of the infinitely large set of heights so close to 164 centimeters that you cannot humanly measure the difference is remote, and thus we assign a probability of 0 to the event.
This is not the case, however, if we talk about the probability of selecting a person who is at least 163 centimeters but not more than 165 centimeters tall.
Now we are dealing with an interval rather than a point value of our random variable.

We shall concern ourselves with computing probabilities for various intervals of continuous random variables such as $P(a < X < b), P(W \geq c)$, and so forth. Note that when X is continuous, $P(a < X \leq b) = P(a < X < b) + P(X = b) = P(a < X < b)$. That is, it does not matter whether we include an endpoint of the interval or not. This is not true, though, when X is discrete.

Probability density function

The function $f(x)$ is a probability density function (pdf) for the continuous random variable X, defined over the set of real numbers, if

- $f(x) \geq 0$, for all $x \in \mathbb{R}$,
- $\int_{-\infty}^{\infty} f(x)dx = 1$.

Example

Suppose that the error in the reaction temperature, in C, for a controlled laboratory experiment is a continuous random variable X having the probability density function

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2 \\ 0. & \text{elsewhere.} \end{cases}$$

1. Verify that $f(x)$ is a density function.
2. Find $P(0 < X \leq 1)$. $= \int_0^1 f(x)\,dx = \int_0^1 \frac{x^2}{3}\,dx = \left[\frac{x^3}{9}\right]_0^1 = 1/9$

clearly $f(x) \geq 0 \quad \forall x \in \mathbb{R}$.

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{-1} f(x)\,dx + \int_{-1}^{2} f(x)\,dx + \int_{2}^{\infty} f(x)\,dx$$

$$= \int_{-1}^{2} \frac{x^2}{3}\,dx = \frac{1}{3}\left[\frac{x^3}{3}\right]_{-1}^{2} = \frac{1}{3}\left[\frac{8+1}{3}\right]$$

$$= 1$$

f is a probability density function.

Cumulative distribution function

The cumulative distribution function $F(x)$ of a random variable X with probability distribution $P(x)$ and probability desity function $f(x)$ is

$$\boxed{F(x) = P(X \leq x) = \begin{cases} \sum_{t \leq x} P(X = t), & \text{If } X \text{ is discrete} \\ \int_{-\infty}^{x} f(x)dx, & \text{If } X \text{ is continuous} \end{cases}}$$

Let $F(x)$ be a cumulative distribution function of a continuous random variable X. Then $P(a < X < b) = F(b) - F(a)$ and $f(x) = \frac{d(F(x))}{dx}$, if the derivative exists.

Problem

A random variable X has the following probability mass function is given

X	-2	-1	0	1	2	3
P(X)	0.1	k	0.2	$2k$	0.3	k

1. Find k.
2. Evaluate $P(X \leq 2)$ and $P(-1 \leq X \leq 2)$.
3. Find the cumulative distribution function.

Soln: we know $(x_i, p(x_i))$ represents a probability mass function

$$\sum_{i=1}^{6} p(x_i) = 1$$

$$p(-2) + p(-1) + p(0) + p(1) + p(2) + p(3) = 1$$

$$0.1 + k + 0.2 + 2k + 0.3 + k = 1$$

$$\boxed{k = 0.1}$$

X	-2	-1	0	1	2	3
p(X)	0.1	0.1	0.2	0.2	0.3	0.1

$$p(X \leq 2) = p(X=-2) + p(X=-1) + p(X=0) + p(X=1) + p(X=2)$$

$$= 0.9$$

$$p(-1 \leq X \leq 2) = p(X=-1) + p(X=0) + p(X=1) + p(X=2)$$

$$= 0.8$$

X	-2	-1	0	1	2	3
p(X)	0.1	0.1	0.2	0.2	0.3	0.1
F(X)	0.1	0.2	0.4	0.6	0.9	1

A random variable X has the following probability mass function is given

X	0	1	2	3	4	5	6	7
P(X)	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

1. Find k.
2. Evaluate $P(X \leq 6)$, $P(X \geq 6)$ and $P((1.5 < X < 4.5)/(X > 2))$
3. Find the cumulative distribution function.

Soln $\sum_{j=0}^{7} p(x=j) = 1$

$0+k+2k+2k+3k+k^2+2k^2+7k^2+k=1$

$10k^2+9k-1=0$

$\boxed{k=\frac{1}{10}}$

x	0	1	2	3	4	5	6	7
p(x)	0	1/10	2/10	2/10	3/10	$\frac{1}{100}$	$\frac{2}{100}$	$\frac{17}{100}$

$$P(1.5<X<4.5 \mid X>2) = \frac{P((1.5<X<4.5)\cap(X>2))}{P(X>2)}$$

$$\boxed{P(A|B) = \frac{P(A\cap B)}{P(B)}}$$

$$= \frac{P(X=3)+P(X=4)}{P(X=3)+P(X=4)+P(X=5)+P(X=6)+P(X=7)}$$

$$= 5/7$$

Problem

Let X be a continuous random variable with probability density function (pdf) is

$$f(x) = \begin{cases} 2x, & 0 < x < 1 \\ 0. & \text{elsewhere.} \end{cases}$$

1. Find $P(X \leq 0.4)$, $P(X \geq \frac{3}{4})$ and $P(X \geq \frac{1}{2}.)$
2. Evaluate $P(\frac{1}{2} < X \leq \frac{3}{4})$ and $P(X > \frac{3}{4}/X > \frac{1}{2})$.
3. Find the cumulative distribution function.

Soln

(1) $P(X \leq 0.4) = \int_0^{0.4} 2x\,dx = [x^2]_0^{0.4} = 0.16$

$P(X \geq 3/4) = \int_{3/4}^{1} 2x\,dx = [x^2]_{3/4}^{1} = 1 - \frac{9}{16} = 7/16$

(2) $P(\frac{1}{2} < X \leq 3/4) = \int_{1/2}^{3/4} 2x\,dx = [x^2]_{1/2}^{3/4} = \frac{9}{16} - \frac{1}{4} = 5/16$

$$P(X > 3/4 \mid X > 1/2) = \frac{P((X > 3/4) \cap (X > 1/2))}{P(X > 1/2)}$$

$$= \frac{P(X > 3/4)}{P(X > 1/2)} = \frac{\int_{3/4}^{1} 2x\,dx}{\int_{1/2}^{1} 2x\,dx}$$

$$= \frac{[x^2]_{3/4}^{1}}{[x^2]_{1/2}^{1}} = 7/12$$

$$F(x) = P(X \leq x) = \int_0^x 2t\,dt = [t^2]_0^x = x^2$$

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ x^2 & \text{if } 0 \leq x \leq 1 \\ 1 & \text{if } x > 1 \end{cases}$$

Pb If the density function of a continuous RV X is given by

$$f(x) = \begin{cases} a x & 0 \le x \le 1 \\ a & 1 \le x \le 2 \\ 3a - a x & 2 \le x \le 3 \\ 0 & \text{elsewhere} \end{cases}$$

(a) Find the value of a

(b) find cdf of X.

Soln we have $\int_{-\infty}^{\infty} f(x) = 1 \Rightarrow \int_0^1 a x\,dx + \int_1^2 a\,dx + \int_2^3 (3a - ax)\,dx = 1$

$$\Rightarrow a\left(\frac{x^2}{2}\right)_0^1 + a(x)_1^2 + \left(3ax - \frac{ax^2}{2}\right)_2^3 = 1$$

$$\Rightarrow \boxed{a = 1/2}$$

If $x < 0$, then $F(x) = P(X \le x) = 0$

$0 \le x \le 1$ then $F(x) = \int_0^x \frac{x}{2}\,dx = x^4/4$

$1 \le x \le 2$ then $F(x) = \int_0^1 \frac{x}{2}\,dx + \int_1^x \frac{1}{2}\,dx = \frac{x}{2} - 1/4$

$2 \le x \le 3$ then $F(x) = \int_0^1 \frac{x}{2}\,dx + \int_1^2 \frac{1}{2} + \int_2^x \left(\frac{3}{2} - \frac{x}{2}\right)dx$

$$= \frac{3}{2}x - \frac{x^2}{4} - 5/4$$

$x > 3$ then $F(x) = 1$

Problem

The Department of Energy (DOE) puts projects out on bid and generally estimates what a reasonable bid should be. Call the estimate b. The DOE has determined that the density function of the winning (low) bid is

$$f(y) = \begin{cases} \frac{5}{8b}, & \frac{2}{5}b \le y \le 2b \\ 0. & \text{elsewhere.} \end{cases}$$

Find $F(y)$ and use it to determine the probability that the winning bid is less than the DOE's preliminary estimate b.